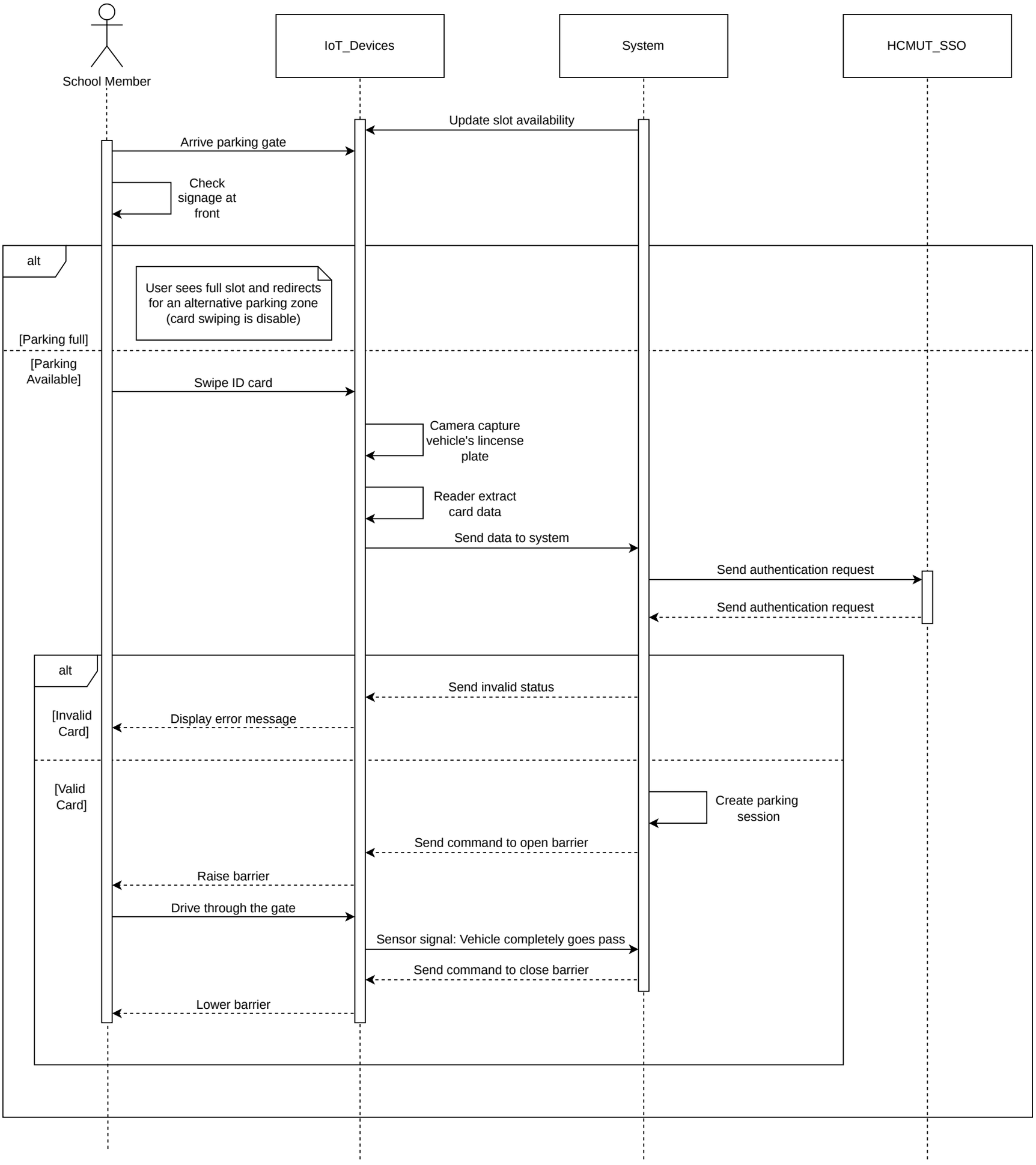
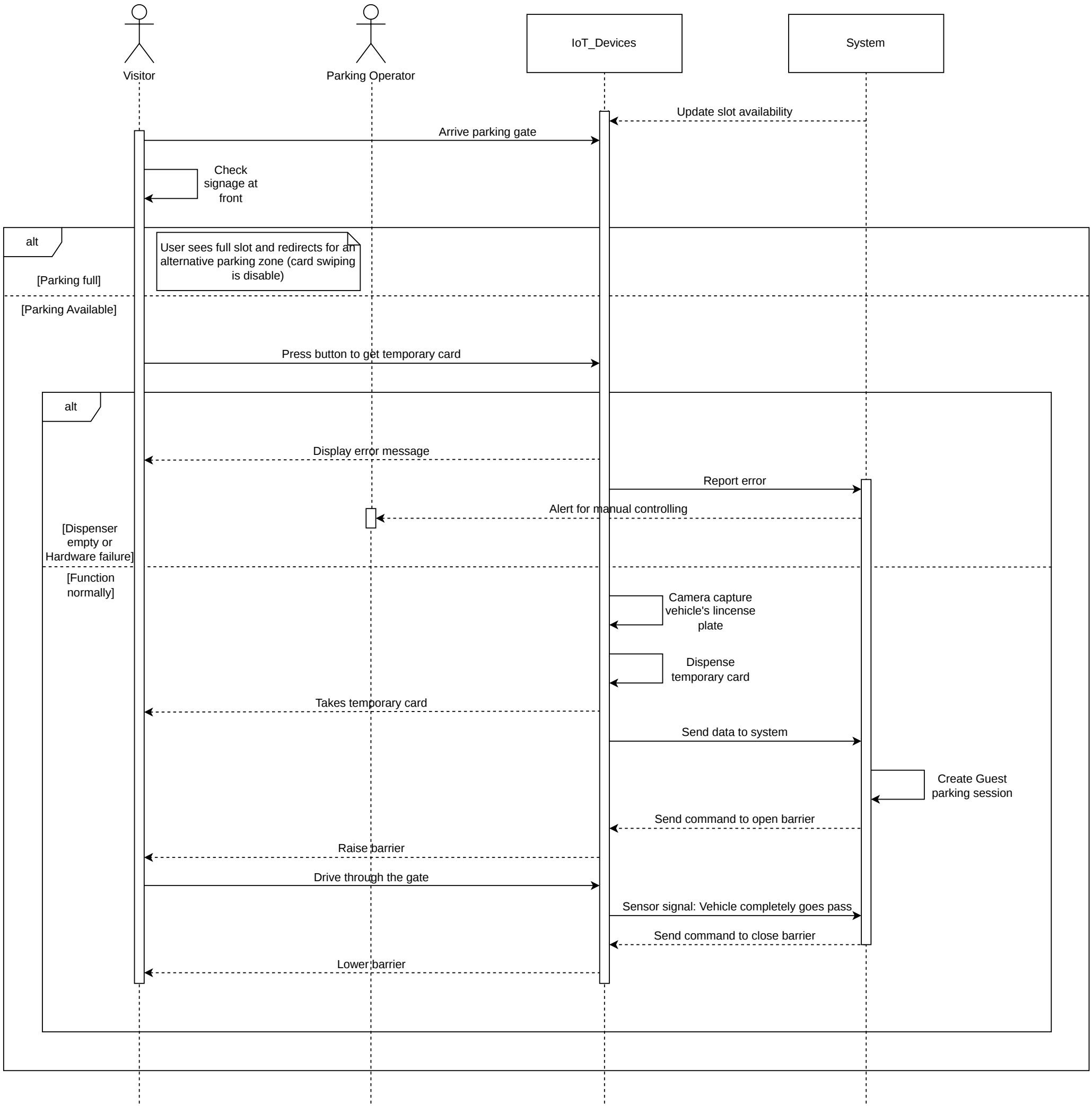
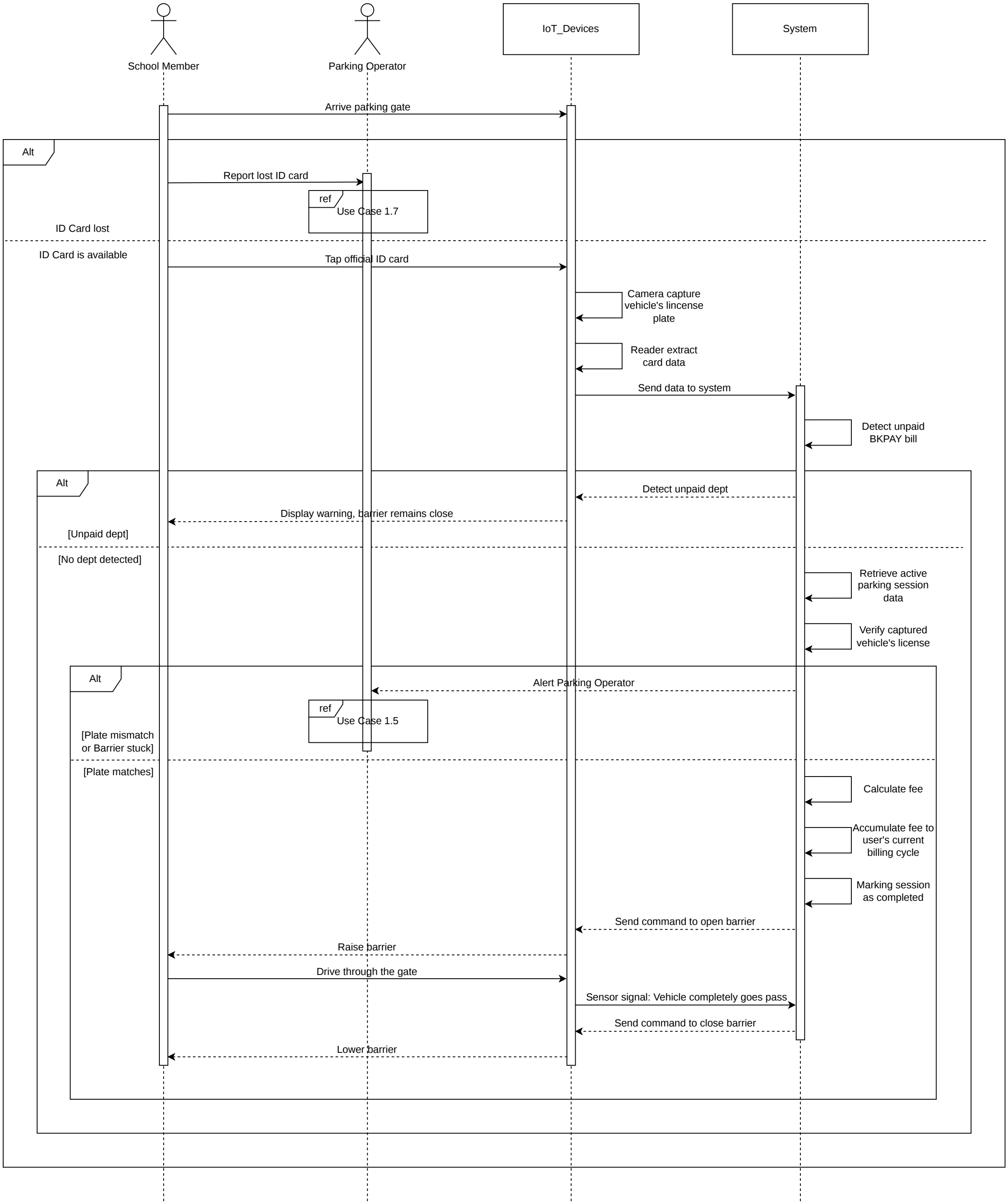
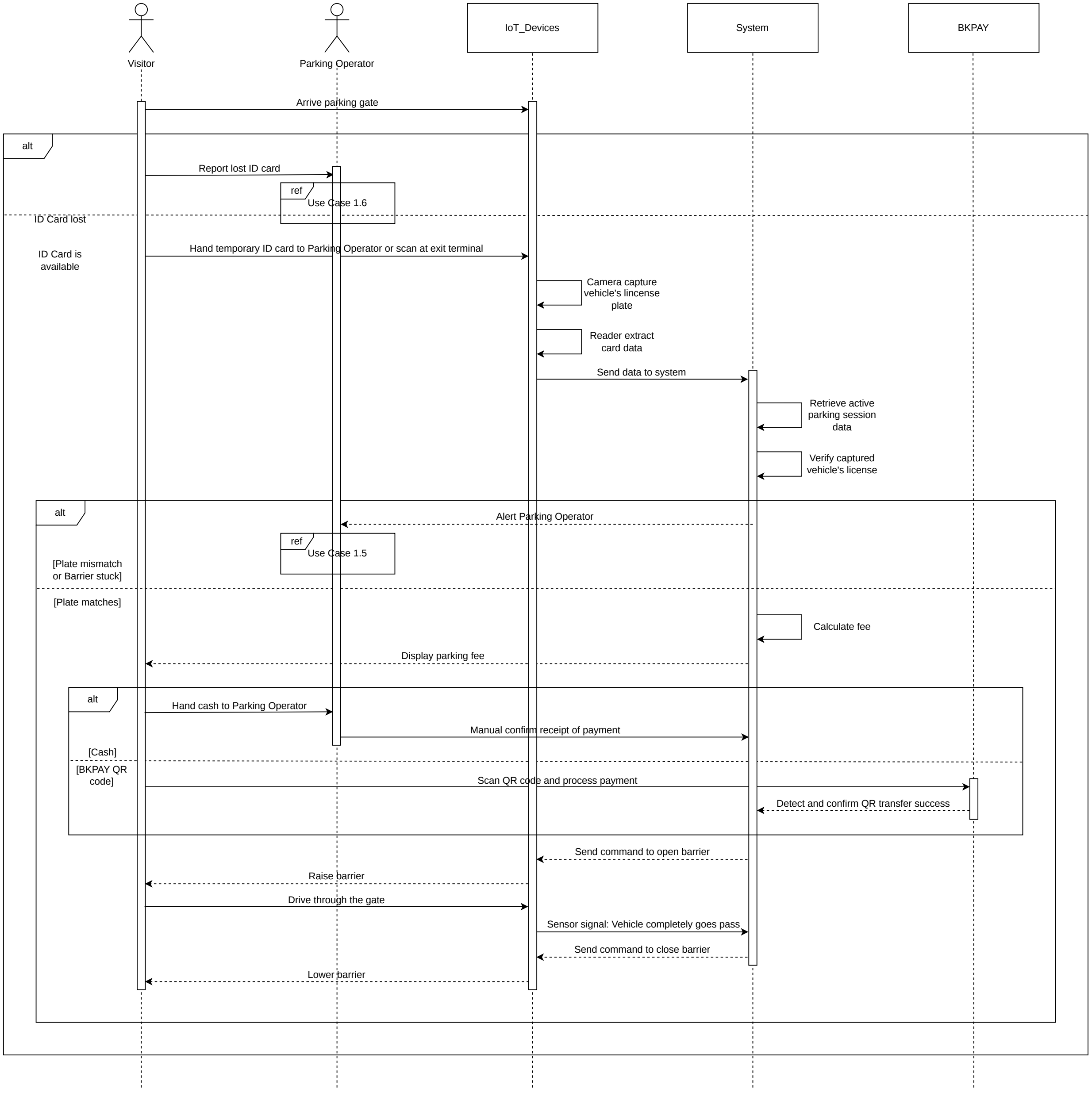
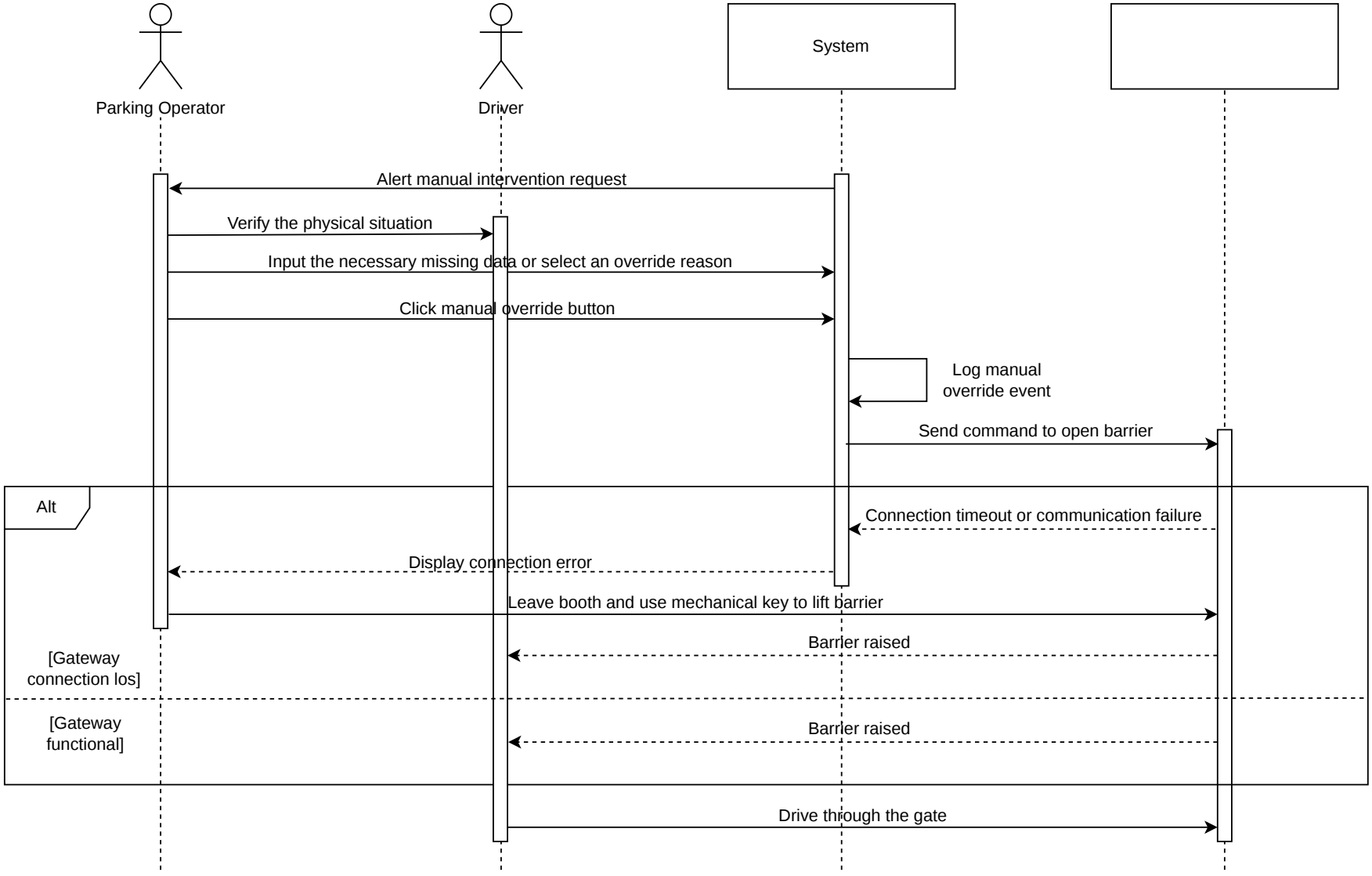
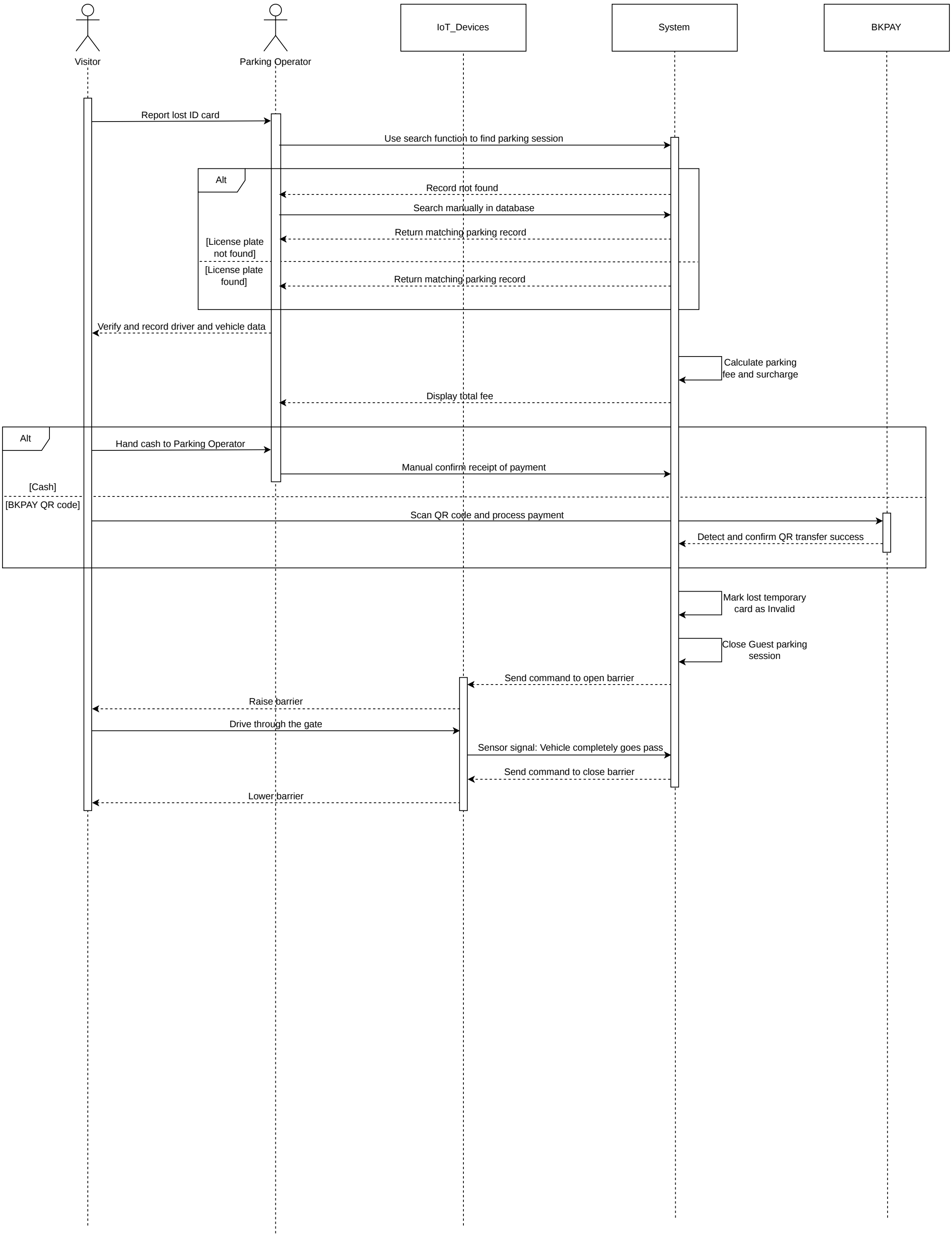

School Member
IoT_Devices
System
HCMUT_SSO
Update slot availability
Arrive parking gate
Check signage at front
alt
User sees full slot and redirects for an alternative parking zone (card swiping is disable)
[Parking full]
[Parking Available]
Swipe ID card
Camera capture vehicle's lincense plate
Reader extract card data
Send data to system
Send authentication request
Send authentication request
alt
Send invalid status
[Invalid Card]
Display error message
[Valid Card]
Create parking session
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier

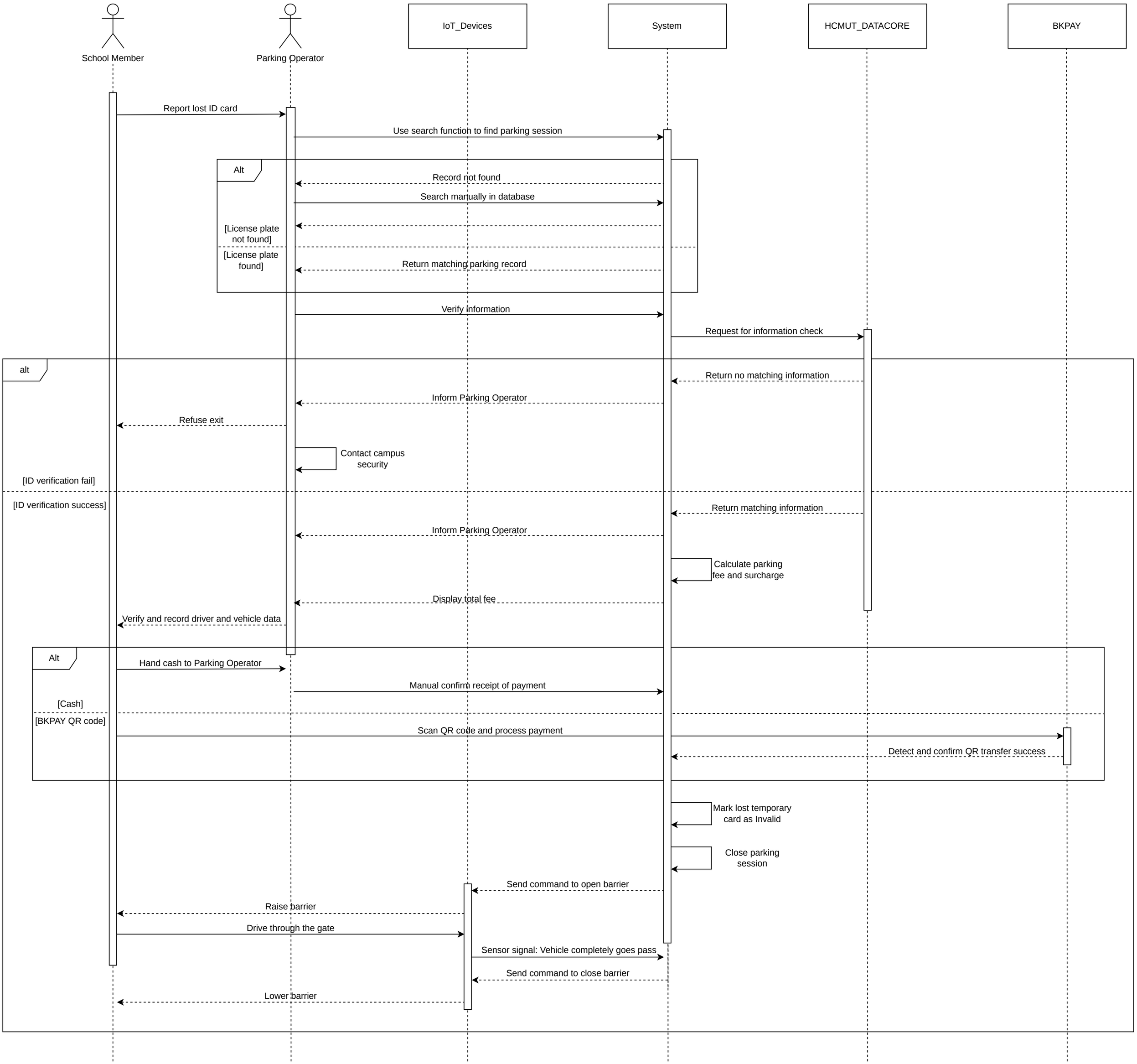

Visitor
Parking Operator
IoT_Devices
System
Update slot availability
Arrive parking gate
Check signage at front
alt
User sees full slot and redirects for an alternative parking zone (card swiping is disable)
[Parking full]
[Parking Available]
Press button to get temporary card
alt
Display error message
Report error
Alert for manual controlling
[Dispenser empty or Hardware failure]
[Function normally]
Camera capture vehicle's lincense plate
Dispense temporary card
Takes temporary card
Send data to system
Create Guest parking session
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier

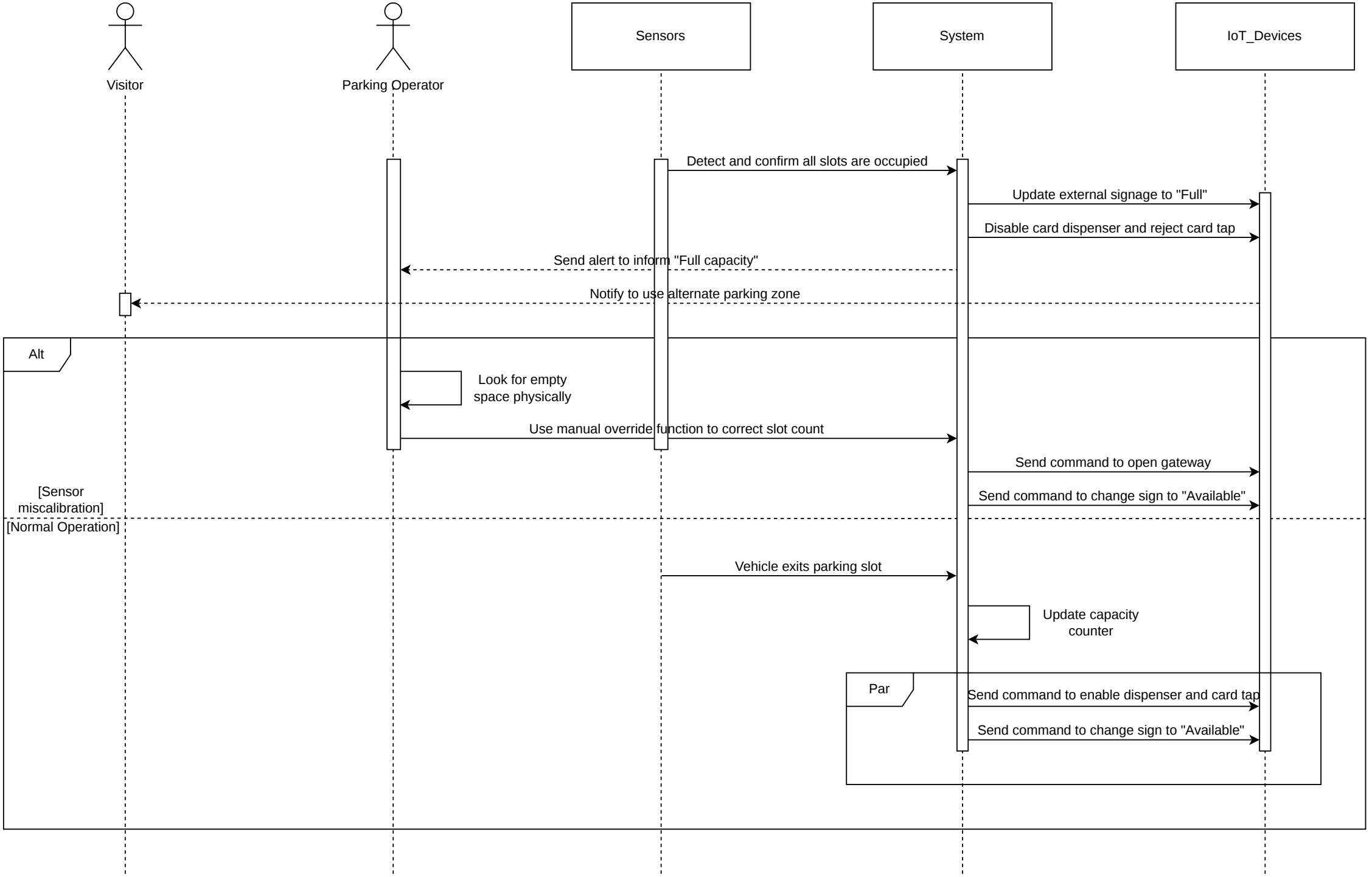

School Member
Parking Operator
IoT_Devices
System
Arrive parking gate
Alt
Report lost ID card
ref
Use Case 1.7
ID Card lost
ID Card is available
Tap official ID card
Camera capture vehicle's lincense plate
Reader extract card data
Send data to system
Detect unpaid BKPAY bill
Alt
Detect unpaid dept
Display warning, barrier remains close
[Unpaid dept]
[No dept detected]
Retrieve active parking session data
Verify captured vehicle's license
Alt
Alert Parking Operator
ref
Use Case 1.5
[Plate mismatch or Barrier stuck]
[Plate matches]
Calculate fee
Accumulate fee to user's current billing cycle
Marking session as completed
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier


Visitor
Parking Operator
IoT_Devices
System
BKPAY
Arrive parking gate
alt
Report lost ID card
ref
Use Case 1.6
ID Card lost
ID Card is available
Hand temporary ID card to Parking Operator or scan at exit terminal
Camera capture vehicle's lincense plate
Reader extract card data
Send data to system
Retrieve active parking session data
Verify captured vehicle's license
alt
Alert Parking Operator
ref
Use Case 1.5
[Plate mismatch or Barrier stuck]
[Plate matches]
Calculate fee
Display parking fee
alt
Hand cash to Parking Operator
Manual confirm receipt of payment
[Cash]
[BKPAY QR code]
Scan QR code and process payment
Detect and confirm QR transfer success
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier


Parking Operator
Driver
System
Alert manual intervention request
Verify the physical situation
Input the necessary missing data or select an override reason
Click manual override button
Log manual override event
Send command to open barrier
Alt
Connection timeout or communication failure
Display connection error
Leave booth and use mechanical key to lift barrier
Barrier raised
[Gateway connection los]
[Gateway functional]
Barrier raised
Drive through the gate


Visitor
Parking Operator
IoT_Devices
System
BKPAY
Report lost ID card
Use search function to find parking session
Alt
Record not found
Search manually in database
Return matching parking record
[License plate not found]
[License plate found]
Return matching parking record
Verify and record driver and vehicle data
Calculate parking fee and surcharge
Display total fee
Alt
Hand cash to Parking Operator
Manual confirm receipt of payment
[Cash]
[BKPAY QR code]
Scan QR code and process payment
Detect and confirm QR transfer success
Mark lost temporary card as Invalid
Close Guest parking session
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier


School Member
Parking Operator
IoT_Devices
System
HCMUT_DATACORE
BKPAY
Report lost ID card
Use search function to find parking session
Alt
Record not found
Search manually in database
[License plate not found]
[License plate found]
Return matching parking record
Verify information
Request for information check
alt
Return no matching information
Inform Parking Operator
Refuse exit
Contact campus security
[ID verification fail]
[ID verification success]
Return matching information
Inform Parking Operator
Calculate parking fee and surcharge
Display total fee
Verify and record driver and vehicle data
Alt
Hand cash to Parking Operator
Manual confirm receipt of payment
[Cash]
[BKPAY QR code]
Scan QR code and process payment
Detect and confirm QR transfer success
Mark lost temporary card as Invalid
Close parking session
Send command to open barrier
Raise barrier
Drive through the gate
Sensor signal: Vehicle completely goes pass
Send command to close barrier
Lower barrier


Visitor
Parking Operator
Sensors
System
IoT_Devices
Detect and confirm all slots are occupied
Update external signage to "Full"
Disable card dispenser and reject card tap
Send alert to inform "Full capacity"
Notify to use alternate parking zone
Alt
Look for empty space physically
Use manual override function to correct slot count
Send command to open gateway
Send command to change sign to "Available"
[Sensor miscalibration]
[Normal Operation]
Vehicle exits parking slot
Update capacity counter
Par
Send command to enable dispenser and card tap
Send command to change sign to "Available"